

3.4.5 Magnetic Fields

- **Magnetic flux density**

Force on a current-carrying wire in a magnetic field.

$F = B I l$, when field is perpendicular to current.

Fleming's left hand rule.

Magnetic flux density B and definition of the tesla.

- **Moving charges in a magnetic field**

Force on charged particles moving in a magnetic field.

$F = B Q v$, when the field is perpendicular to velocity.

Circular path of particles; application in devices such as the cyclotron.

- **Magnetic flux and flux linkage**

Magnetic flux defined by $\Phi = BA$, where B is normal to A .

Flux linkage as $N\Phi$, where N is the number of turns cutting the flux.

Flux and flux linkage passing through a rectangular coil rotated in a magnetic field:

flux linkage $N\Phi = BAN \cos \theta$ where θ is the angle between the normal to the plane of the coil and the magnetic field.

- **Electromagnetic induction**

Simple experimental phenomena.

Faraday's and Lenz's laws.

Magnitude of induced emf = rate of change of flux linkage = $N \frac{\Delta \phi}{\Delta t}$

Applications such as a moving straight conductor.

Emf induced in a coil rotating uniformly in a magnetic field:

$$E = BAN\omega \sin \omega t$$

The operation of a transformer;

$$\text{The transformer equation} = \frac{N_s}{N_p} = \frac{V_s}{V_p}$$

$$\text{Transformer efficiency} = I_s V_s / I_p V_p$$

Causes of inefficiency of a transformer.

Transmission of electrical power at high voltage.